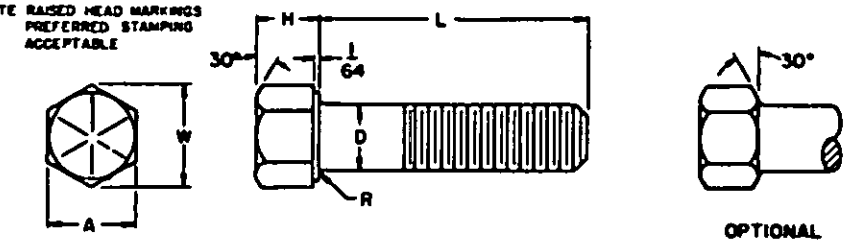
REVIEWER: N1, EL, MR, VC, SH
USER: OS, MC, CL, AT, MU

FED. SUP CLASS 5305

NOTE RAISED HEAD MARKINGS PREFERRED STAMPING ACCEPTABLE



NOMINAL SIZE			1/4	5/16	3/8	7/16	1/2	9/16	5/8
THREADS PER INCH			28UNF	24UNF	24UNF	20UNF	20UNF	18UNF	18UNF
BODY DIAMETER	D	Max	.2500	.3125	.3750	.4375	.5000	.5625	.6250
		Min	.2450	.3065	.3690	.4305	.4930	.5545	.6170
WIDTH ACROSS FLATS	A	Max	.4375	.5000	.5625	.6250	.7500	.8125	.9375
		Min	.4250	.4890	.5510	.6120	.7360	.7980	.9220
WIDTH ACROSS CORNERS	W	Max	.5060	.5770	.6500	.7220	.8660	.9380	1.0830
		Min	.4890	.5570	.6240	.6980	.8400	.9100	1.0510
HEAD HEIGHT	H	Max	.1630	.2110	.2430	.2910	.3230	.3710	.4010
		Min	.1500	.1960	.2260	.2720	.3020	.3480	.3750
RADIUS	R	Max	.023	.023	.023	.023	.023	.041	.041
		Min	.009	.009	.009	.009	.009	.021	.021

LENGTH	L	Tolerance	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/8			-4	-5	-6				
7/16			-5	-6	-7	-8			
1/2			-6	-7	-8	-9	-10		
9/16			-7	-8	-9	-10	-11	-12	
5/8		+0 -1/32	-8	-9	-10	-11	-12	-13	-14
3/4			-10	-11	-12	-13	-14	-15	-16
7/8			-11	-12	-13	-14	-15	-16	-17
1			-12	-13	-14	-15	-16	-17	-18
1-1/8			-13	-14	-15	-16	-17	-18	-19
1-1/4			-14	-15	-16	-17	-18	-19	-20
1-3/8		+0 -1/16	-15	-16	-17	-18	-19	-20	-21
1-1/2			-16	-17	-18	-19	-20	-21	-22
1-3/4			-18	-19	-20	-21	-22	-23	-24
2			-20	-21	-22	-23	-24	-25	-26
2-1/4			-22	-23	-24	-25	-26	-27	-28
2-1/2			-24	-25	-26	-27	-28	-29	-30
2-3/4			-26	-27	-28	-29	-30	-31	-32
3			-28	-29	-30	-31	-32	-33	-34
3-1/4			-30	-31	-32	-33	-34	-35	-36
3-1/2		+0 -3/32	-32	-33	-34	-35	-36	-37	-38
3-3/4			-34	-35	-36	-37	-38	-39	-40
4			-36	-37	-38	-39	-40	-41	-42
4-1/4			-38	-39	-40	-41	-42	-43	-44
4-1/2			-40	-41	-42	-43	-44	-45	-46
4-3/4			-42	-43	-44	-45	-46	-47	-48
5			-44	-45	-46	-47	-48	-49	-50
5-1/2							-52	-53	-54
6							-56	-57	-58

MATERIAL: Alloy Steel, Grade 5 of procurement specification. Minimum tensile strengths: 150,000 psi for diameters through 1-1/2"; 125,000 psi for diameters over 1-1/2".
FINISH: Plain (untreated).
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.
MINIMUM THREAD LENGTH: Shall be twice the diameter plus 1/4" for lengths up to and including 6". The tolerance shall be plus 1/16" or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, for sizes up to and including 1" and 3-1/2 threads for sizes larger than 1".
NOTES: (1) Referenced documents of issue in effect on date of invitation for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example: MS35302-1.
(4) All dimensions in inches.

CANCELLED IN PART
SEE SHEET 2

© For change see sheet 1.

APPROVED MARCH 4, 1954 REVISED (A) 8 MAY 63 (B) 9 JUNE 1967 (C) 14 JUN 1971

This military standard has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. VC	TITLE	MILITARY STANDARD
Other Cust SH	SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, PLAIN FINISH, UNF-2A	MS 35302
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES	SHEET 1 OF 2

DD FORM 672-1 (Continued)

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FED. SUP CLASS
5305

REVIEWER: MI, EL, ME, VC, SH
USER: OS, MC, CL, AT, MU

NOMINAL SIZE			3/4	7/8	1	1-1/8	1-1/4	1-3/8	1-1/2
THREADS PER INCH			16UNF	14UNF	12UNF	12UNF	12UNF	12UNF	12UNF
BODY DIAMETER	D	Max	.7500	.8750	1.0000	1.1250	1.2500	1.3750	1.5000
		Min	.7410	.8660	0.9900	1.1140	1.2390	1.3630	1.4880
WIDTH ACROSS FLATS	A	Max	1.1250	1.3125	1.5000	1.6875	1.8750	2.0625	2.2500
		Min	1.1000	1.2950	1.4690	1.6310	1.8120	1.9940	2.1750
WIDTH ACROSS CORNERS	W	Max	1.2990	1.5160	1.7320	1.9490	2.1650	2.3820	2.5980
		Min	1.2540	1.4650	1.6750	1.8590	2.0660	2.2730	2.4800
HEAD HEIGHT	H	Max	.4830	.5630	.6270	.7180	.8130	.9780	.9740
		Min	.4550	.5310	.5910	.6580	.7490	.8100	.9020
RADIUS	R	Max	.041	.062	.093	.093	.093	.093	.093
		Min	.021	.041	.062	.062	.062	.062	.062

LENGTH L	TOLERANCE Size 3/4 and under	TOLERANCE Size 7/8 and over	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/8	+0 -1/32	+0 -1/16	~~179~~						
7/8			~~180~~	~~092~~					
1			~~181~~	~~093~~	~~224~~				
1-1/8			~~182~~	~~094~~	~~225~~	~~245~~			
1-1/4			~~183~~	~~095~~	~~226~~	~~246~~	~~265~~		
1-3/8	+0 -1/16	+0 -1/8	~~184~~	~~096~~	~~227~~	~~247~~	~~266~~	~~284~~	
1-1/2			~~185~~	~~097~~	~~228~~	~~248~~	~~267~~	~~285~~	~~302~~
1-3/4			~~186~~	~~098~~	~~229~~	~~249~~	~~268~~	~~286~~	~~303~~
2			~~187~~	~~099~~	~~230~~	~~250~~	~~269~~	~~287~~	~~304~~
2-1/4			~~188~~	~~210~~	~~231~~	~~251~~	~~270~~	~~288~~	~~305~~
2-1/2			~~189~~	~~211~~	~~232~~	~~252~~	~~271~~	~~289~~	~~306~~
2-3/4			~~190~~	~~212~~	~~233~~	~~253~~	~~272~~	~~290~~	~~307~~
3			~~191~~	~~213~~	~~234~~	~~254~~	~~273~~	~~291~~	~~308~~
3-1/4			~~192~~	~~214~~	~~235~~	~~255~~	~~274~~	~~292~~	~~309~~
3-1/2			~~193~~	~~215~~	~~236~~	~~256~~	~~275~~	~~293~~	~~310~~
3-3/4	+0 -3/32	+0 -3/16	~~194~~	~~216~~	~~237~~	~~257~~	~~276~~	~~294~~	~~311~~
4			~~195~~	~~217~~	~~238~~	~~258~~	~~277~~	~~295~~	~~312~~
4-1/4			~~196~~	~~218~~	~~239~~	~~259~~	~~278~~	~~296~~	~~313~~
4-1/2			~~197~~	~~219~~	~~240~~	~~260~~	~~279~~	~~297~~	~~314~~
4-3/4			~~198~~	~~220~~	~~241~~	~~261~~	~~280~~	~~298~~	~~315~~
5			~~199~~	~~221~~	~~242~~	~~262~~	~~281~~	~~299~~	~~316~~
5-1/2			~~200~~	~~222~~	~~243~~	~~263~~	~~282~~	~~300~~	~~317~~
6			~~201~~	~~223~~	~~244~~	~~264~~	~~283~~	~~301~~	~~318~~

NOTE: For illustration, material, finish and other pertinent data, see sheet 1.

CANCELLED IN PART AFTER 9 June 1967. USE MS90727. INTERCHANGEABILITY RELATIONSHIP - MS90727 can replace the cancelled screws of the same size. The cancelled MSJ5302 cannot always replace MS90727 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

CANCELLED IN PART
SEE NOTE

APPROVED MARCH 4, 1954 REVISED (A) 6 MAY 63 (B) 9 JUN 1967 (C) FOR CHANGES SEE SHEET 1.

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Other Cust SH	SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, PLAIN FINISH, UNF-2A	MS 35302
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 2 OF

DD FORM 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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